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Evaluating Safety Metrics for Vulnerable Road Users at Urban Traffic Intersections Using High-Density Infrastructure LiDAR System

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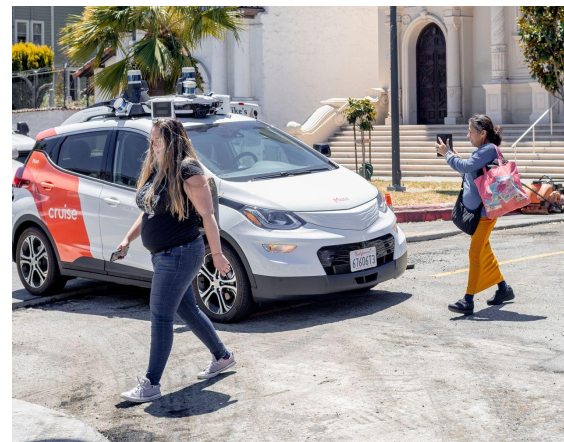
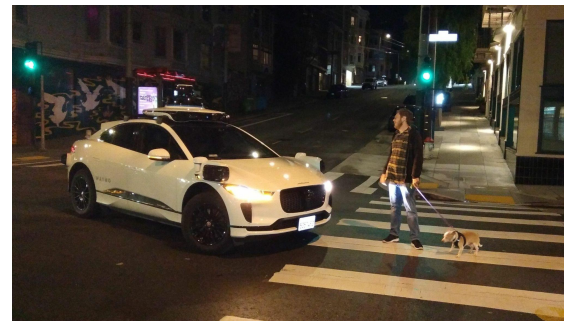


Institute of
Automated Mobility



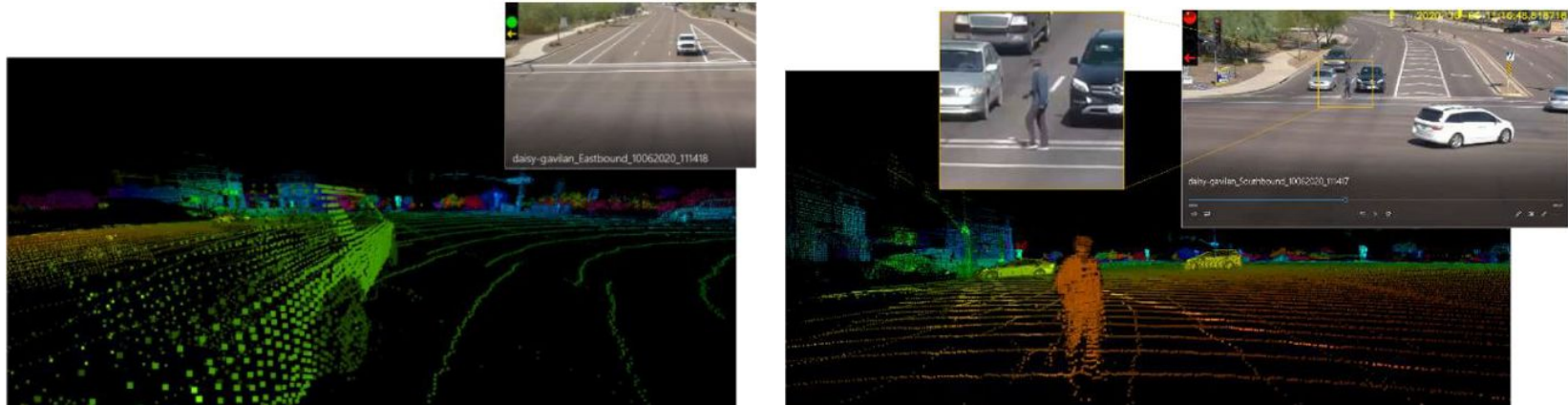
Traffic Safety with Infrastructure Sensors

- Over 50% of traffic-related fatalities and injuries occur at intersections, which requires automated traffic safety infrastructure at high-risk zones.
- Aggressive driving behavior and distracted driving causes many drivers to collide with pedestrians and other vehicles in traffic negotiation.
- A major goal for safety is to use sensors that can perceive vehicle position and velocities while providing safety to drivers in real-time.



Introduction – Using LiDAR Data for Traffic Safety

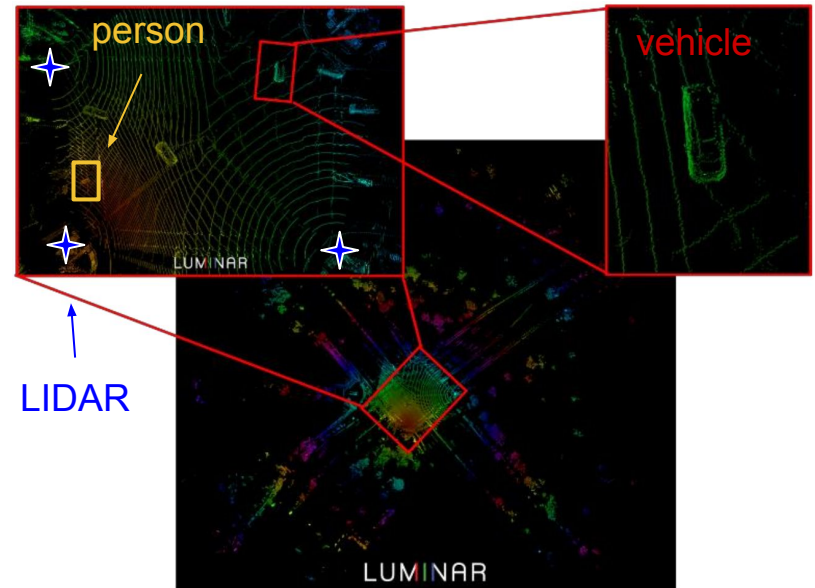
- Traffic safety infrastructure has incorporated the use of modern technologies such as LiDAR for real-time vehicle detection.



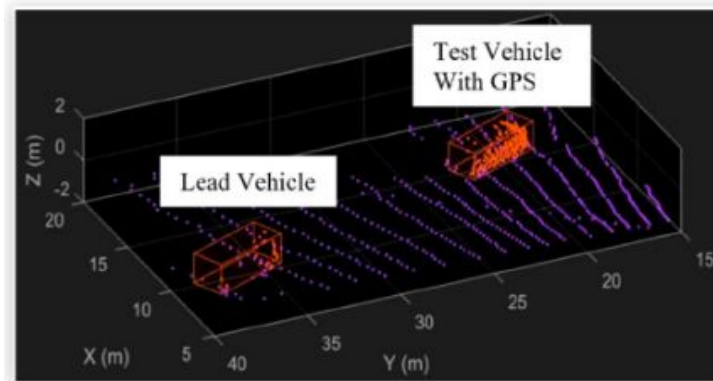
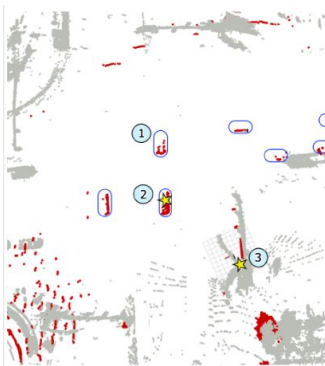
Niraj Altekar, Steven Como, Duo Lu, Jeffrey Wishart, Donald Bruyere, Faisal Saleem, and K. Larry Head. **"Infrastructure-based Sensor Data Capture Systems for Measurement of Operational Safety Assessment (OSA) Metrics."** *SAE International Journal of Advances and Current Practices in Mobility* 3, no. 2021-01-0175 (2021): 1933-1944.

Introduction – Advantages of LiDAR Data

- With a reported centimeter level error and sampling rate of nearly 1.5 million points per second, LiDAR data have high resolution and accuracy.
- From point cloud data, objects may be detected using current SOTA deep learning models and tracked using bayesian filters.



Prior Work – LiDAR-based Safety Metrics Computation



1. From prior work made by Anshuman *et al.* [1], showed position and velocity measurements from LiDAR may be used to calculate operational safety metrics.
2. Follow up work from Siddarth *et al.* [2] compared fidelity between infrastructure-based and vehicle-top-based LiDAR setups for metrics computation.
3. Elli *et al.* [3] used Minimum Distance Safety Envelope (MDSE) safety metrics for vehicle-to-vehicle interactions and compared it with conventional metrics such as Post Encroachment Time (PET).

Introduction – Objectives

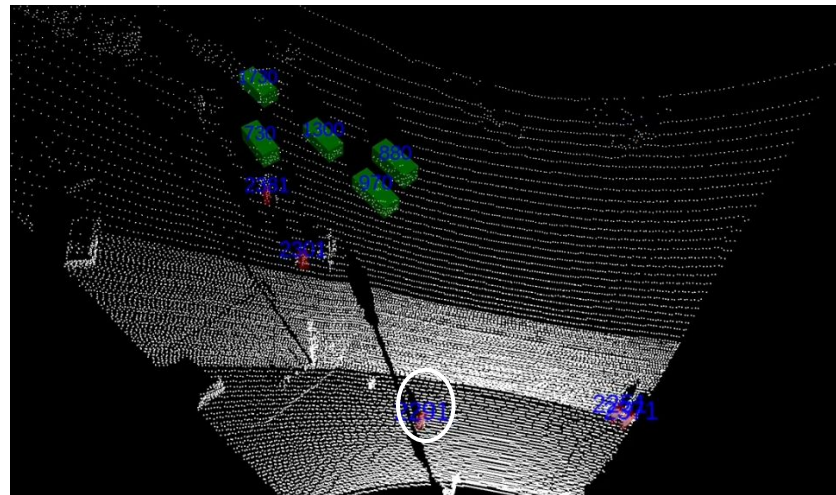
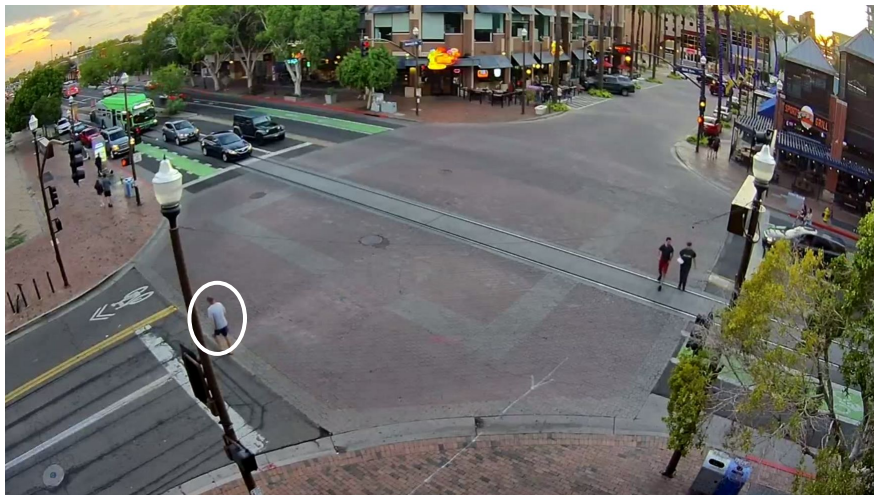
In this work, we focus on Vulnerable Road Users (VRUs):

- Pedestrians
- Bicyclists
- Micro Mobility Vehicles

Objectives of our work

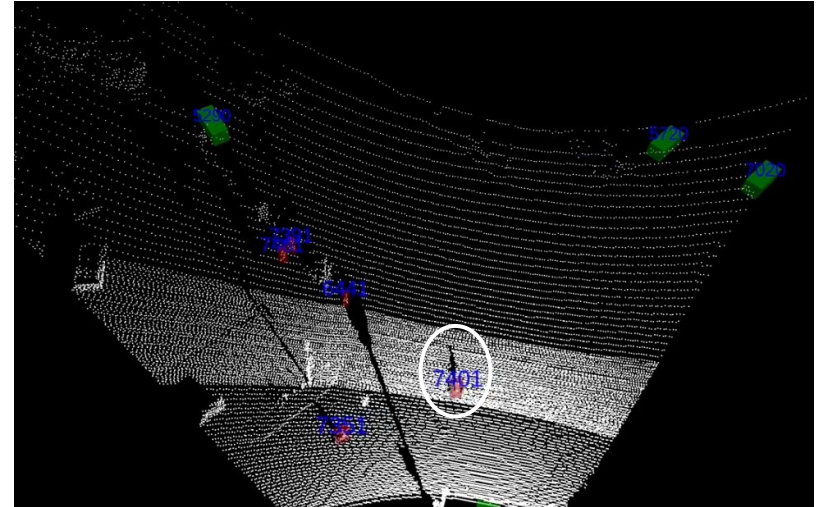
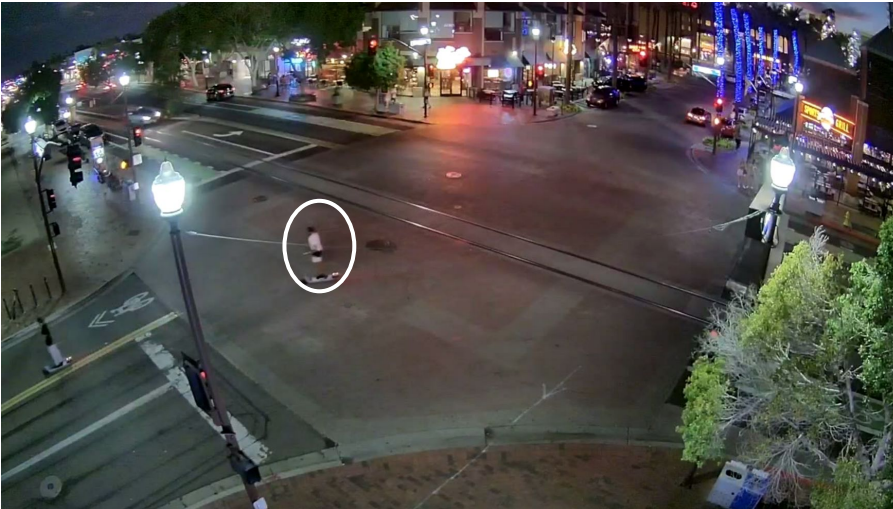
1. Understand the dynamics of traffic for real-time situational awareness.
2. Predict potentially unsafe situations for VRUs and vehicles.
3. Evaluate real-time safety metrics related to vehicles and VRUs.

Pedestrian Detection and Tracking



Day time pedestrian detection and tracking

Micro-Mobility Detection and Tracking



Night time micro-mobility vehicle (e-scooter) detection and tracking

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Prior works only consider local and highway intersections with only vehicles.

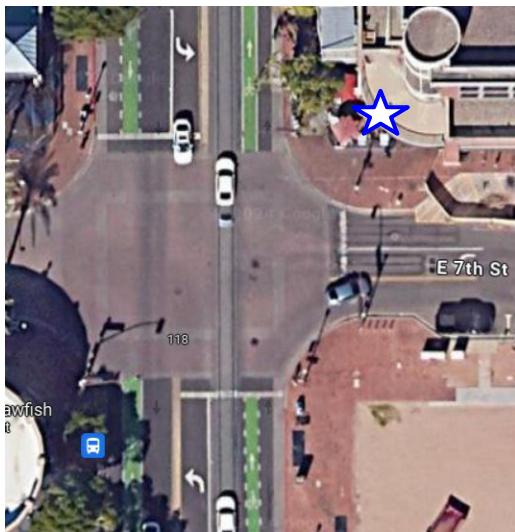
We extend this work to urban traffic situations for VRU safety analysis.

We intend to answer the following key questions

1. Are infrastructure-based LiDARs suitable for analyzing urban traffic scenarios?
2. How can we compute safety metrics for analyzing interactions between vehicles and VRUs?

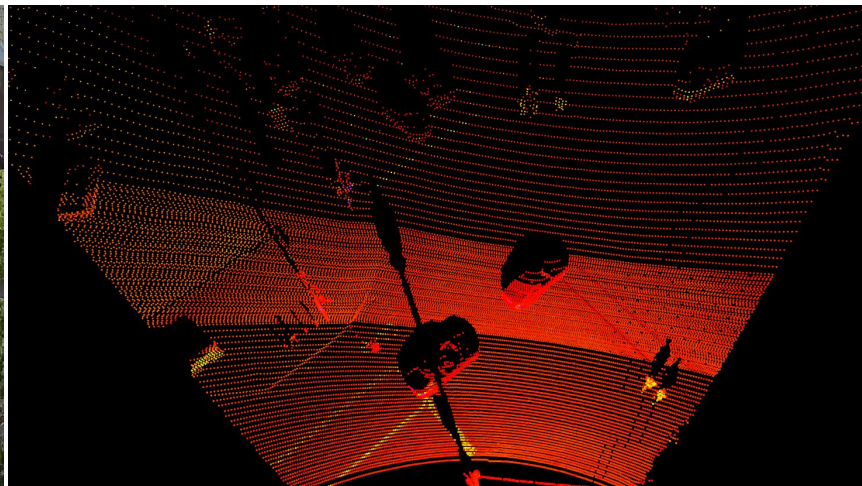
Data Collection

For our analysis we collect data from an urban traffic intersection at testbed located in downtown Tempe, Arizona



Data Collection

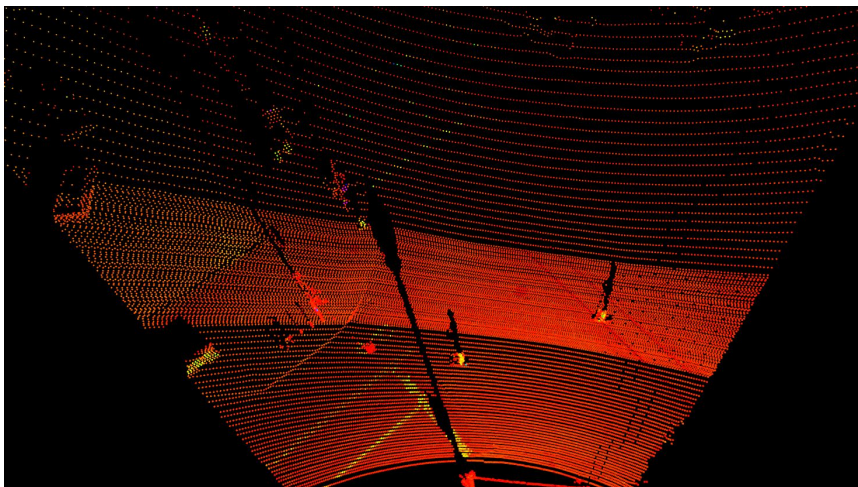
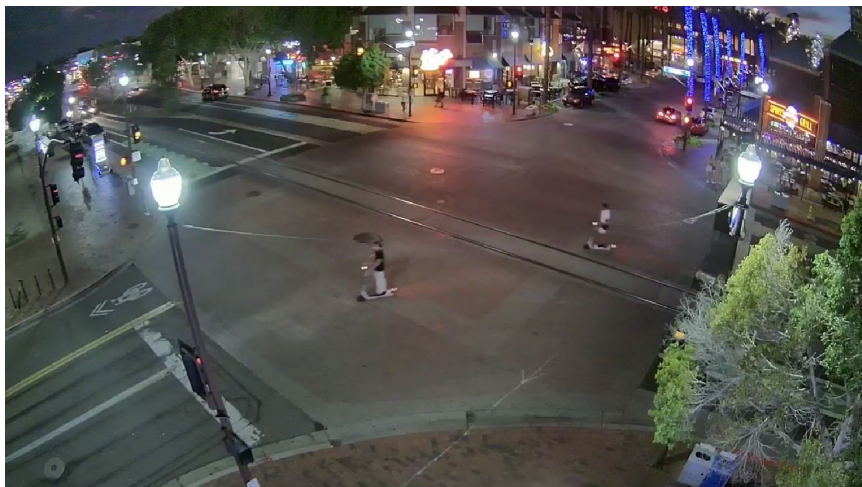
We collect data during the dusk period, with the objective of capturing the transition from daylight to night.



Day time Data Collection

Data Collection

We collect data during the dusk period, with the objective of capturing the transition from daylight to night.



Night time Data Collection

Data Collection

- Four batches of data at different time of the day are collected.
- In total, we collected 18 minutes of traffic data.

Interval #	Start Time	End Time
1	6:28 pm	6:30 pm
2	6:38 pm	6:41 pm
3	7:08 pm	7:11 pm
4	7:13 pm	7:23 pm

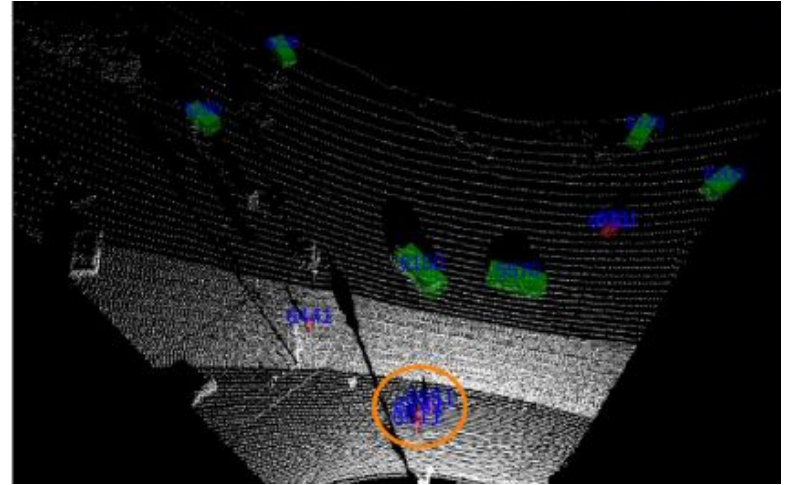
Data Collection Batches

Methodology - Detection and Tracking

We use a PV-RCNN [4] model pretrained on KITTI dataset for 3D object detection and Kalman Filters for tracking.

1. High density 128 channel LiDARs capture dense VRU features that are essential for reliable detection.
2. The pointcloud is downsampled using 10 cm voxel grid to ensure uniform point density.
3. Different Kalman Filters are used for vehicles and VRUs as we empirically find the optimal parameters to be highly dependent on object class.

Methodology - Detection and Tracking

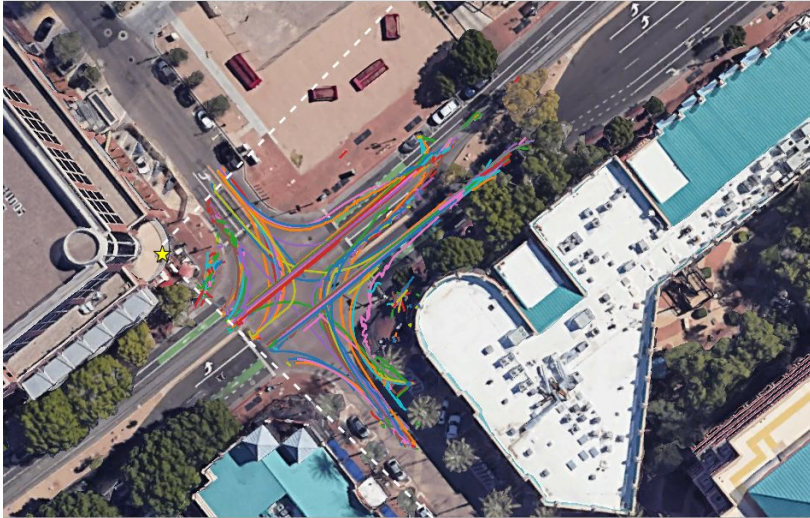


Object Detection Performance - LiDAR vs Camera

Table below shows the Precision (P) and Recall (R) values for both Camera and LiDAR detections during day and night time. LiDAR shows more reliability.

Day Time - Camera				Day Time - LiDAR			
	Ped	Veh	All		Ped	Veh	All
P	0.434	0.656	0.545	P	0.876	0.991	0.933
R	0.380	0.698	0.539	R	0.532	0.798	0.665
Night Time - Camera				Night Time - LiDAR			
	Ped	Veh	All		Ped	Veh	All
P	0.143	0.494	0.377	P	0.827	0.944	0.885
R	0.384	0.756	0.632	R	0.761	0.791	0.776

Methodology - Detection and Tracking

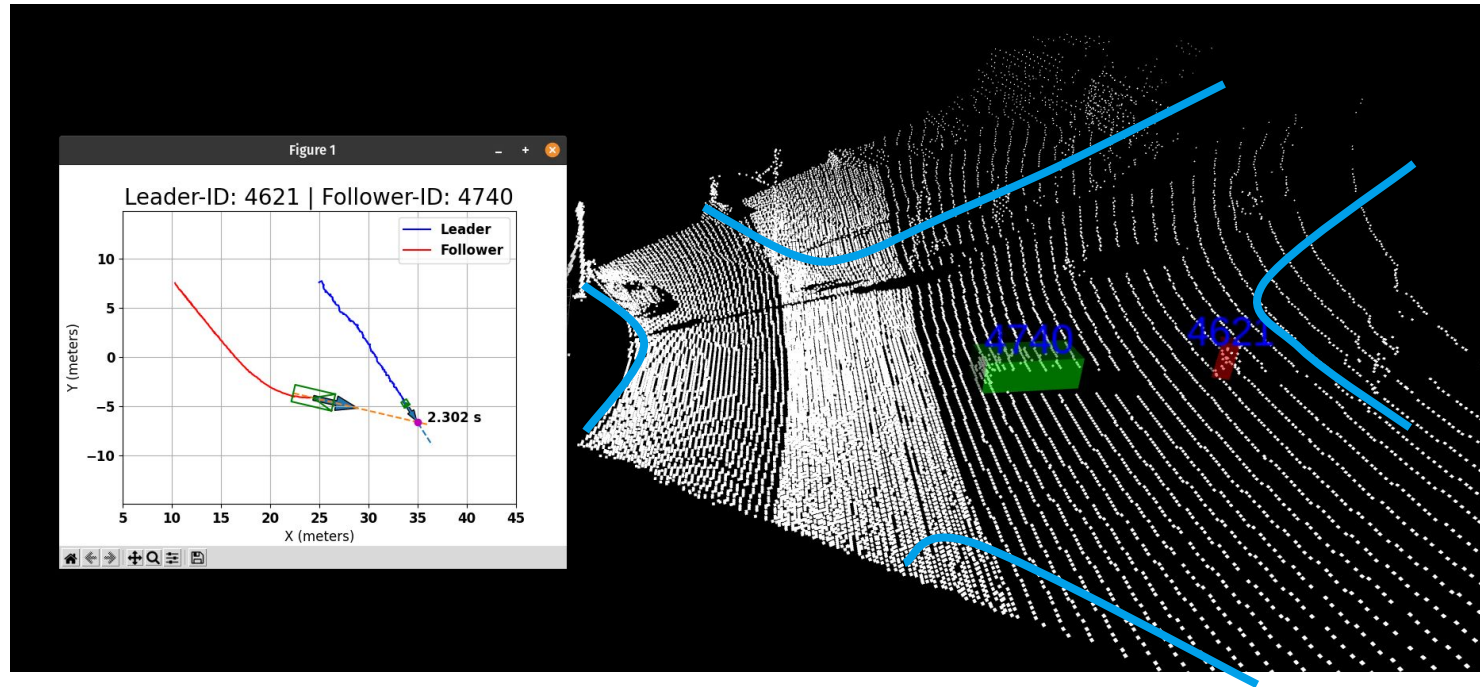


(left) Vehicle tracklets and (right) VRU tracklets over a period of 1 hour from our tracking pipeline. Overlaid on a Google Earth image of the Tempe intersection.

Methodology - Identifying Unsafe Interactions

- We use the real-time tracking information to identify potentially unsafe interactions between detected objects.
 - We extrapolate the vehicle trajectory using the instantaneous velocity vector times 5 seconds into the future.
 - 2D vector to vector intersections are computed using Bentley-Ottmann algorithm [5] to determine real-time conflict points.
- All entities involved in such interactions are uniquely identified to be in unsafe situations.

Methodology - Safety Metrics



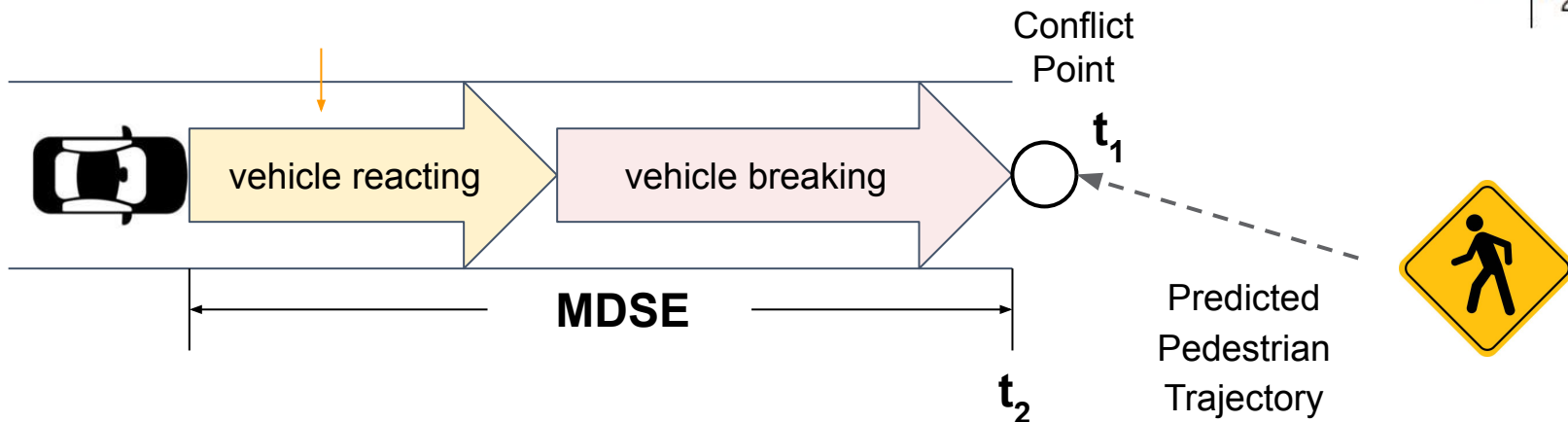
The pink conflict point is the intersection between scaled velocity vectors.

- For every unsafe pair in the intersection, we evaluate real-time safety metrics with respect to the detected conflict point.
 - Post Encroachment Time (PET)
 - Minimum Distance Safety Envelope (MDSE)
- Time complexity of our approach is reduced compared to exhaustive pairwise checking in prior works.
- Our approach is faster and more suitable for real-time computation.

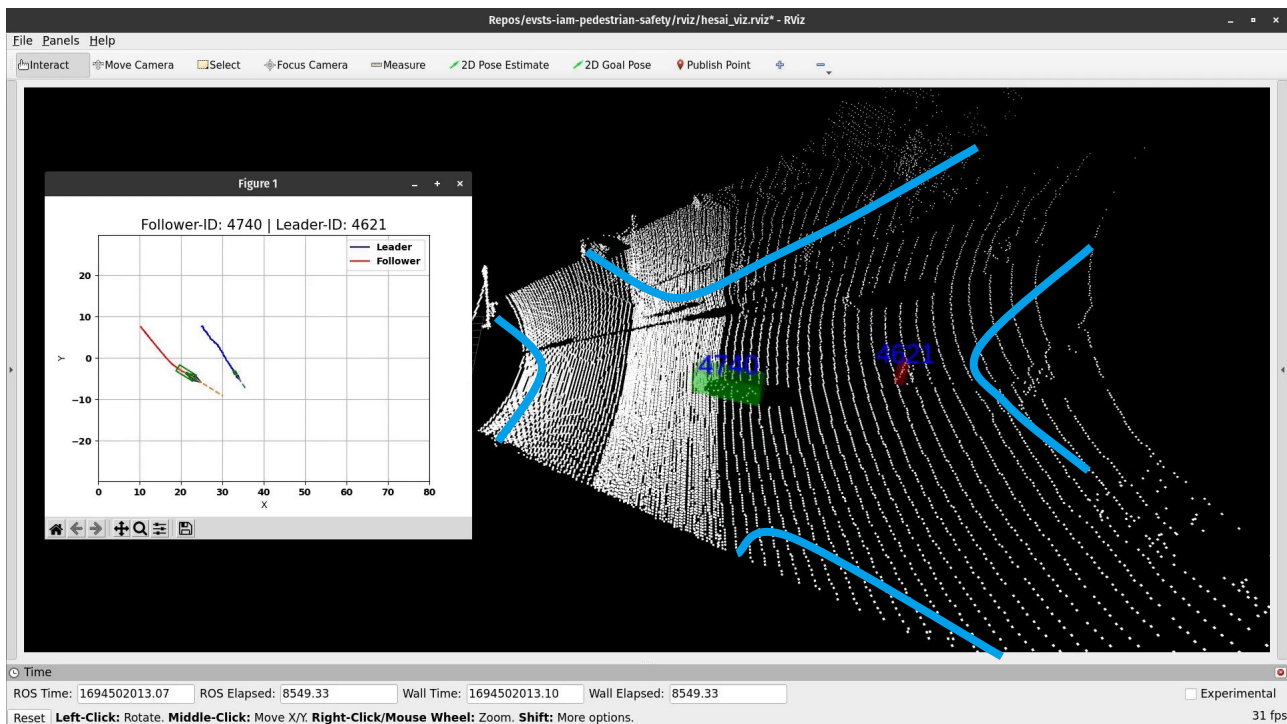
Methodology - Safety Metrics

$$\text{MDSE} = v_l^{\text{long}} \rho_l + \frac{1}{2} a_{l,\text{max,accel}}^{\text{long}} \rho_l^2 + \frac{\left(v_l^{\text{long}} + \rho_l a_{l,\text{max,accel}}^{\text{long}} \right)^2}{2 a_{l,\text{min,decel}}^{\text{long}}}$$

$$\text{PET} = |t_2 - t_1|$$

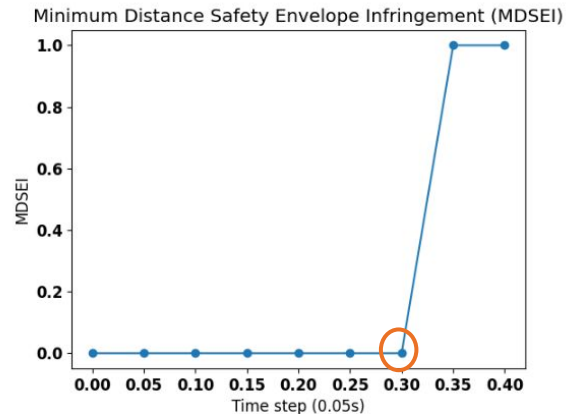
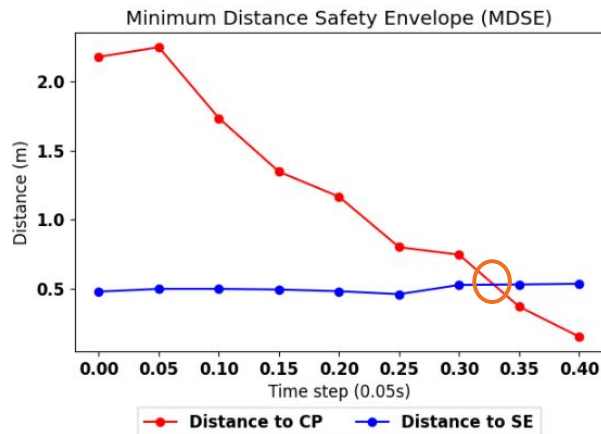
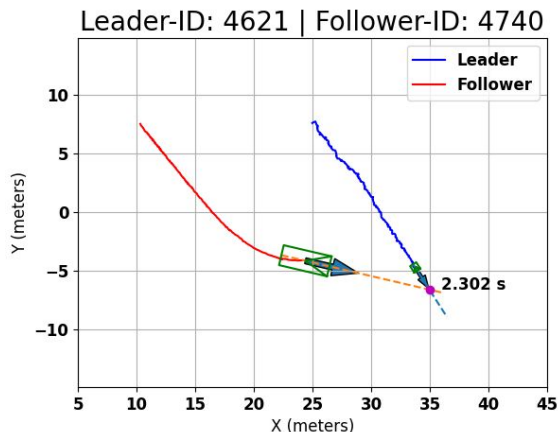


Methodology - Safety Metrics



VRU-to-vehicle unsafe situation with real-time PET metrics calculation.

Safety Violations



- A PET violation occurs when the PET value drops below 1.5 seconds.
- A MDSE violation occurs when the distance to the conflict point drops within the calculated safety envelope.

Results - Safety Violations

Table below shows the safety metrics violation for VRU-vehicle interactions.

Interval #	VRU track-lets	Unsafe situations	PETV	MDSEV	Common violations
1	23	5	2	0	0
2	12	5	1	0	0
3	27	12	8	4	3
4	147	37	27	15	10
Total	232	59	38	19	13

Conclusion

Our work underscores the importance of real-time and continuous monitoring systems for VRU safety for urban traffic scenarios.

Our contributions:

1. An LiDAR-based infrastructure monitoring system for detecting potentially unsafe situations.
2. Analysis on the effectiveness of DA metrics for safety of VRUs.
3. Open-source data for the AV community for VRU safety research.

References

- [1] A. Srinivasan, Y. Mahartayasa, V. Jammula, D. Lu, S. Como, J. Wishart, Y. Yang, and H. Yu, “Infrastucture-Based LIDAR Monitoring for Assessing Automated Driving Safety,” SAE Technical Paper 2022-01-0081, 2022.
- [2] S. Das, P. Rath, D. Lu, T. Smith, J. Wishart, and H. Yu, “Comparison of Infrastructure-and Onboard Vehicle-Based Sensor Systems in Measuring Operational Safety Assessment (OSA) Metrics,” tech. rep., SAE Technical Paper, 2023.
- [3] M. S. Elli, J. Wishart, S. Como, S. Dhakshinamoorthy, and J. Weast, “Evaluation of operational safety assessment (osa) metrics for automated vehicles in simulation,” tech. rep., SAE Technical Paper, 2021.
- [4] S. Shi, C. Guo, L. Jiang, Z. Wang, J. Shi, X. Wang, and H. Li, “Pv-rcnn: Point-voxel feature set abstraction for 3d object detection,” in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10529–10538, 2020.
- [5] Bentley and Ottmann, “Algorithms for reporting and counting geometric intersections,” IEEE Transactions on computers, vol. 100, no. 9, pp. 643–647, 1979.